

WYVERN-E: Engineering Design and Experimental Validation of Closed-Loop Thrust-Vector Control, Hybrid Passive-Active Stability, and Zoned Additive-Manufacturing Materials in a Subscale Single Stage Solid-Fuelled Prototype Rocket Demonstrator

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Abstract

The WYVERN-E vehicle is a 70 mm-diameter, single-stage, solid-fuelled, additively-manufactured prototype rocket demonstrator developed to provide quantitative, ground-based and flight-validated data on closed-loop thrust-vector control (TVC) actuation, hybrid passive/active stability architectures, and zone-specific additive-manufacturing material selection. The vehicle is powered by a single Estes F15-0 solid motor and relies on four fixed fins sized to the minimum statically-stable margin (1.0 caliber) to survive launch rail departure and the motor's ignition transient, after which authority is transferred to a closed-loop proportional-integral-derivative (PID) thrust-vector-control system operating on the smooth portion of the thrust curve. Two candidate TVC actuation schemes — a tri-solenoid magnetic gimbal and a servo-actuated gimbal — are characterized and directly compared on a purpose-built three-axis thrust-vector load balance prior to flight, isolating actuator dynamics (bandwidth, slew rate, overshoot, steady-state error, and maximum achievable deflection angle) from flight-to-flight variability. All flight avionics, sensor fusion, and control-law execution are consolidated onto a single dual-core microcontroller (Raspberry Pi Pico 2 W, RP2350), with one core dedicated exclusively to deterministic 500 Hz control execution and the second core handling data logging and wireless telemetry — eliminating the blocking-I/O risk inherent to single-threaded flight computer architectures. Structural members are fabricated by fused deposition modeling (FDM) using a thermally-zoned material strategy: a flame-and-heat-rated polycarbonate (PC-FR) in the motor and gimbal bays, and a low-density foamed acrylonitrile styrene acrylate (ASA-Aero) elsewhere, reducing dry mass by approximately 100–150 g relative to a uniform-material baseline. Aerofoil sections are evaluated in a custom open-return low-speed wind tunnel constructed to the Hofferth (2025) modular design, providing ground-based aerodynamic characterization that is cross-validated against flight telemetry. All flights are conducted under FAA Class 1 (model rocket) provisions, requiring no airworthiness waiver. Pre-flight Monte Carlo and validation simulation predict an apogee near 432 ft (132 m) with a positive thrust-to-weight ratio throughout the burn (2.07 average, 3.62 peak) and a closed-loop pitch deviation under 1° across the modeled atmospheric envelope. All CAD, firmware, simulation code, and flight and ground-test datasets will be released publicly through a version-controlled, openly accessible Git repository and presented in summary form within this paper upon completion of the program.

1. Introduction

1.1 Background and Motivation

Active flight control in small-scale, hobbyist-accessible rocketry sits at the intersection of aerodynamics, embedded real-time systems, additive-manufacturing materials science, and classical control theory, and progress on any one axis is frequently constrained by the others. Thrust-vector control — gimbaling the rocket motor's nozzle or exhaust path to produce a control moment without dedicated aerodynamic control surfaces — is the actuation method used on essentially every orbital launch vehicle, yet open, reproducible, and quantitatively validated implementations at the sounding-rocket scale remain comparatively rare in the amateur and academic literature. Existing low-cost demonstrations of vector control (e.g., BPS.space's *Signal/Echo* flight series) establish feasibility but do not typically isolate and quantify actuator-class performance independent of flight-to-flight aerodynamic and atmospheric variability.

WYVERN-E addresses this gap with a deliberately simplified architecture relative to prior program iterations: a single fixed-fin airframe that is passively stable through the motor's ignition transient, a single bare-metal flight controller executing the entire sensing-and-control pipeline, and — critically — an actuator comparison that is moved off the flight vehicle and onto a repeatable ground-test apparatus. This restructuring is intended to produce a controlled, statistically tractable dataset on actuator performance (the central engineering question of the program) while retaining a flight-validation phase that demonstrates the complete closed-loop system performs as designed under real atmospheric and motor-burn conditions.

1.2 Program Lineage

The WYVERN-E line has progressed through four major design iterations. WYVERN-E 1.0 was a 70 mm, two-stage airframe with a custom flight computer, magnetic-solenoid thrust-vector-control actuation, and actively controlled fins on both stages; 2.0 introduced an 84 mm two-stage airframe with custom avionics; 3.0 implemented a Raspberry Pi 5-based flight computer and flew a magnetic-solenoid-versus-servo actuator A/B comparison in flight. Each iteration established components of the aerodynamic, structural, and avionics groundwork carried forward into the current design. The current iteration consolidates these lessons into the simplest vehicle configuration that still answers the program's central control-authority question, replacing the in-flight actuator A/B test with a ground-based load-balance comparison and replacing the distributed avionics stack with a single dual-core microcontroller.

1.3 Central Hypothesis

A small fixed-fin, single-stage, solid-fuelled prototype rocket demonstrator can rely on passive aerodynamic stability to survive launch-rail departure and the motor's ignition transient, then transfer control authority to a closed-loop thrust-vector-control system once the thrust curve smooths; and the relative performance of competing TVC actuator classes (magnetic-solenoid versus servo) can be characterized quantitatively, and more

economically, on a ground-based three-axis load balance prior to committing either actuator to flight.

2. Research Questions

Research Question 1 — Magnetic-Solenoid versus Servo Actuation for Thrust-Vector Control Authority. Substituting a tri-solenoid magnetic gimbal actuator for a servo-driven gimbal actuator as the thrust-vector-control mechanism, evaluated under identical commanded-deflection inputs on a three-axis thrust-vector load balance, will produce measurable differences in actuation bandwidth, slew rate, step-response overshoot, steady-state deflection error, and maximum achievable gimbal angle, with the objective of determining which actuator class provides superior control authority for a given motor thrust regime.

Research Question 2 — Structural Performance of Zone-Specific 3D-Printable Materials. Holding all FDM process parameters constant across flame-retardant polycarbonate (PC-FR) and foamed acrylonitrile styrene acrylate (ASA-Aero), this study characterizes and compares each material's flexural stiffness under three-point bend loading, heat-deflection margin under simulated motor-bay thermal loading, and mass-specific structural performance, with the objective of validating a thermally-zoned material allocation (heat-rated polymer confined to the motor/gimbal bay, lightweight foamed polymer elsewhere) as superior to a single-material airframe on a mass-to-margin basis.

Research Question 3 — Aerofoil Profile Efficiency for Passive Subsonic Stability. Testing the fixed fin's symmetric aerofoil cross-section across incrementally increasing angles of attack in subsonic wind-tunnel flow, lift coefficient, drag coefficient, lift-to-drag ratio, and stall onset angle will be measured and compared against thin-airfoil theory and published low-Reynolds-number reference data, with the objective of confirming that the selected fin geometry provides sufficient restoring moment to maintain the minimum stable static margin through the pre-TVC flight phase without imposing an unnecessary drag or mass penalty.

Research Question 4 — Wind-Tunnel Calibration Against In-Situ Flight Performance. Incrementally adjusting fan speed across a range of test-section velocities in the custom open-return low-speed wind tunnel, the Reynolds-number similarity ratio, blockage-corrected drag coefficient, and cross-sectional flow uniformity will be evaluated at each condition to identify the operating point at which tunnel-measured aerodynamic loading most closely reproduces the loading predicted on the full-scale vehicle during powered ascent, and tunnel-derived drag coefficients will be cross-validated against telemetry-derived drag estimates from flight.

Research Question 5 — Closed-Loop Control Gain Sensitivity in a Single-Controller TVC Architecture. Operating the proportional-integral-derivative thrust-vector control loop on the single flight microcontroller under multiple candidate gain sets (including a step-response-tuned baseline and a simulation-refined gain set validated across modeled atmospheric and gust conditions), peak pitch deviation, gimbal-angle utilization, and

settling behavior will be quantified and compared across repeated flights to determine which gain configuration produces the most accurate and best-damped trajectory tracking on a deterministic single-core-equivalent control architecture.

3. Project Goal

The goal of the WYVERN-E program is to design, fabricate, ground-test, and flight-validate a recoverable, FAA Class 1, 70 mm-diameter prototype rocket demonstrator that serves as an integrated experimental platform for: (i) quantitatively comparing thrust-vector-control actuator classes independent of flight-to-flight variability; (ii) validating a thermally-zoned additive-manufacturing material strategy against airframe mass and structural-margin objectives; (iii) characterizing the aerodynamic performance of the vehicle's passive-stability fin geometry; (iv) validating a custom low-speed wind tunnel against in-situ flight telemetry; and (v) quantifying control-loop gain sensitivity on a consolidated single-controller avionics architecture. Testing is conducted across three complementary tiers — computational simulation, ground-based instrumented testing, and in-situ powered flight — so that each research question is addressed through at least two independent and mutually cross-validating methods.

4. Vehicle Architecture

4.1 Configuration and Mass Budget

The vehicle is a 70 mm outer-diameter, single-stage airframe approximately 0.74 m in length, comprising three pressure/equipment bays (engine/TVC, flight-computer, recovery) separated by two structural bulkheads, with four fixed fins providing passive stability. The recalculated mass budget, derived from the thermally-zoned material allocation described in §4.2 and Research Question 2, is summarized in Table 1.

Table 1. WYVERN-E mass budget.

Section	Material	Key contents	Mass
Nose cone	PC-FR	—	40 g
Engine/TVC bay	PC-FR	Bulkhead A, gimbal assembly, 2× servo actuators, attitude IMU, motor mount	152 g
Flight-computer bay	ASA-Aero	Pico 2 W, attitude IMU, environmental/baro metric sensors, microSD logger, 2S LiPo, BEC, action	55 g

Recovery bay	ASA-Aero	camera RRC3+ altimeter, 9 V lithium battery, parachute and shock cord, redundant attitude IMU, Bulkhead B	118 g
Structure	ASA-Aero (body) / PC-FR (fins)	Body tube, 4× fixed fins, internal wiring	203 g
Dry mass total			606 g
Motor (loaded)	—	Estes F15-0, 60 g propellant	102 g
Liftoff mass total			708 g

The zoned allocation of foamed ASA-Aero to bays without direct motor-plume thermal exposure, reserving the heat- and flame-rated PC-FR for the motor/gimbal bay, fins, and nose, reduces dry mass by approximately 100–150 g relative to an all-PC-FR baseline (which was specified at 812 g liftoff in an earlier design iteration), directly raising the thrust-to-weight ratio and predicted apogee.

4.2 Stability Architecture: Passive Margin with Active Handoff

Because the closed-loop TVC system cannot be engaged instantaneously at ignition (servo and control-loop settling, combined with the high-amplitude transient of the motor's ignition spike, would risk a destabilizing initial command), the vehicle must be passively stable through approximately the first 0.5 s of flight. Four fixed fins (root chord 70 mm, tip chord 35 mm, leading-edge sweep 25°, span 58 mm) are sized to the minimum conventionally stable static margin of 1.0 caliber at liftoff (center of gravity at 46.7 cm from the nose, center of pressure at 52.5 cm, by the Barrowman (1967) method), increasing to approximately 1.3 caliber by burnout as propellant mass is consumed. A parametric apogee-versus-ballast sweep (Table 2) confirmed that adding nose ballast to increase margin is counter-productive: each gram of ballast lowers apogee, so no ballast is carried, and stability is achieved purely through fin sizing at the minimum stable margin.

Table 2. Apogee sensitivity to nose ballast (RK4 + Barrowman simulation, 1.0 cal margin held constant by fin resizing).

Ballast

0 g (selected)

60 g

150 g

At $t = 0.5$ s, with the motor on the smooth, sustained portion of its thrust curve, the TVC controller is enabled and assumes full attitude authority, commanding the gimbal to

stabilize the vehicle to vertical and subsequently execute a small commanded pitch maneuver, while the fins continue to provide passive restoring moment as a stability backstop for the remainder of the powered phase.

5. Proposed Methodology

The program employs a three-tier experimental framework spanning computational simulation, ground-based instrumented testing, and in-situ powered flight. Each research question is addressed through at least two of these tiers to enable cross-validation. All fabrication uses hobbyist-accessible manufacturing methods and commercially available components, consistent with the program's open-source objectives.

5.1 Airframe and Materials (Research Question 2)

All primary structural components — body tube, nose cone, fin set, and bulkheads — are fabricated via fused deposition modeling (FDM) on a Bambu Lab X1C printer using the stock 0.4 mm brass nozzle. Print parameters (layer height, infill pattern and density, wall count, nozzle and bed temperature, and part cooling) are held constant within each material class to isolate material as the independent variable, following the parameter-control methodology of Popescu et al. (2018). Two candidate materials are evaluated: flame-retardant polycarbonate (PC-FR, $\rho \approx 1.25 \text{ g/cm}^3$, heat-deflection temperature $\approx 110^\circ\text{C}$, UL94-V0 rated), allocated to the nose, engine/TVC bay, structural bulkhead A, and fins where direct or conducted motor-plume heating is expected; and a foamed acrylonitrile styrene acrylate (ASA-Aero, $\rho \approx 0.5\text{--}0.7 \text{ g/cm}^3$), allocated to the body tube and the flight-computer and recovery bays, where no significant thermal load is present. Flexural stiffness of both materials is characterized under standardized three-point bend loading following the mechanical-characterization framework of Dizon et al. (2018). Engine-bay thermal performance is verified by a first-order lumped-capacitance transient model of the PC-FR motor-bay wall against the F15-O's 3.45 s burn (with a 0.5 mm phenolic liner as the inner thermal barrier), predicting a peak wall temperature near 47°C — well below the PC-FR heat-deflection limit — and first-order structural margins on the airframe exceed a safety factor of $300\times$ against the motor's peak axial and TVC-induced bending loads, confirming that the airframe wall thickness is set by printability and handling robustness rather than flight-load requirements.

5.2 Fins and Aerofoil Profile (Research Question 3)

The flight fin employs a symmetric aerofoil cross-section sized to deliver the minimum 1.0 caliber static margin identified in §4.2. Reference low-angle-of-attack lift and drag behavior for symmetric sections of this class is benchmarked against the classical thin-airfoil and low-Reynolds-number frameworks of Lissaman (1983) and Mueller and DeLaurier (2003), and against tabulated NACA section data (Abbott & Von Doenhoff, 1959). Wind-tunnel-measured lift coefficient, drag coefficient, lift-to-drag ratio, and stall onset angle for the fabricated fin article are compared directly against these references to confirm that the selected geometry produces adequate restoring moment without excess parasitic drag, and

the resulting tunnel-measured aerodynamic coefficients feed the Barrowman center-of-pressure calculation used to confirm static margin in §4.2.

5.3 Flight Computer, Sensing, and Control Architecture (Research Questions 1 and 5)

Consolidated single-controller architecture. All flight avionics functions — attitude estimation, control-law execution, actuator commanding, data logging, and telemetry — are consolidated onto a single Raspberry Pi Pico 2 W (RP2350: dual-core 150 MHz Arm Cortex-M33, 520 KB SRAM, on-board Wi-Fi/BLE radio), replacing the distributed multi-board avionics architecture used in earlier program iterations. The two processor cores are functionally partitioned to preserve hard real-time determinism: Core 0 executes the 500 Hz thrust-vector-control loop exclusively — reading the gimbal- and body-mounted inertial measurement units, computing nozzle deflection, evaluating the PID control law, and commanding the gimbal servos — and is permitted no blocking operations of any kind. Core 1 drains a logged-data ring buffer to a microSD card over SPI and services an optional Wi-Fi telemetry link for ground-station monitoring, isolating all non-deterministic I/O latency from the control path. This division directly addresses the principal failure mode of single-threaded flight-computer architectures, in which storage or radio I/O can transiently block control-loop execution.

Attitude sensing. Three nine-axis inertial measurement units (Bosch BNO085) are deployed — one rigidly referenced to the gimbal/nozzle, one to the vehicle body in the flight-computer bay, and a third redundant unit in the recovery bay for two-of-three fault voting — each configured in Game Rotation Vector mode (accelerometer-gyroscope fusion with the magnetometer disabled), because the magnetic field generated by the adjacent gimbal servos would otherwise corrupt a magnetically-referenced heading estimate. Effective nozzle deflection relative to the vehicle body is computed each control cycle as the quaternion difference $q_{\text{defl}} = q_{\text{body}}^{-1} \otimes q_{\text{gimbal}}$, which captures true mechanical nozzle attitude (including any linkage backlash or structural flex) rather than an assumed commanded angle.

Control law (Research Question 5). The per-axis control law is a discrete PID controller with integral anti-windup clamping and a first-order low-pass-filtered derivative term, executed at 500 Hz with output clipped to a $\pm 5^\circ$ gimbal deflection limit. Two candidate gain sets are evaluated empirically: a classical step-response-tuned baseline following the Ziegler–Nichols (1942) method, and a simulation-refined gain set ($K_p = 2.0$, $K_i = 0.4$, $K_d = 0.5$) derived from closed-loop simulation across four atmospheric conditions (cold, ISA-standard, hot, and high-density-altitude) with mid- and late-burn wind-gust disturbances, which is predicted to hold pitch deviation to $0.33\text{--}0.44^\circ$ with gimbal utilization under 2° (against a $\pm 5^\circ$ hardware limit) across the full modeled atmospheric envelope, while avoiding the resonance against finite servo lag (≈ 40 ms) observed in higher-proportional-gain configurations during simulation. The TVC loop is inhibited for the first 0.5 s of flight (§4.2), after which it engages to stabilize the vehicle to vertical and execute a small commanded maneuver; required gimbal torque is estimated at $0.56 \text{ kg}\cdot\text{cm}$, well within the selected servo class, and simulated control authority remains positive throughout the powered phase across the modeled gain sets.

Actuator comparison (Research Question 1). Two TVC actuator classes are evaluated using the identical control electronics, gimbal mechanism, and software control law, isolating actuator dynamics as the experimental variable: a tri-solenoid magnetic gimbal actuator, and a servo-driven gimbal actuator. The comparison is conducted entirely on the ground-based three-axis thrust-vector load balance described in §5.4, rather than in flight, to remove flight-to-flight aerodynamic and atmospheric variability from the actuator comparison; the flight vehicle itself carries the servo actuator, selected on the basis of the ground-comparison results.

5.4 Ground Test Program (Research Question 1)

The ground test program is structured as the program's primary experimental apparatus rather than a preliminary check, since both the actuator-comparison (Research Question 1) and the motor-characterization data that feed every downstream simulation (Research Question 5; §4) are generated here rather than in flight. Two purpose-built stands are constructed: a three-axis thrust-vector load balance for actuator comparison, and a single-axis static-thrust stand for motor-curve verification and materials erosion screening. Both stands are fully printable in PC-FR (selected per Research Question 2 for its motor-plume thermal margin) and instrumented with strain-gauge load cells and HX711 24-bit bridge-amplifier breakouts, logged to onboard microSD by a dedicated data-acquisition microcontroller independent of the flight avionics.

Three-axis thrust-vector load balance. The actuator under test (magnetic-solenoid or servo gimbal) is mounted to a thrust block restrained from a fixed base by three strain-gauge load cells acting through flexures — one axial and two lateral — resolving the complete thrust vector in magnitude and direction:

$$T = \sqrt{F_x^2 + F_y^2 + F_z^2}, \theta = \arctan\left(\frac{\sqrt{F_x^2 + F_y^2}}{F_z}\right), \phi = \text{atan2}(F_y, F_x)$$

The cells are sized to the expected loading envelope of the test motor (Estes F15-0, 25.3 N peak axial thrust, side force at the $\pm 5^\circ$ gimbal limit ≈ 2.2 N), using a 5 kg axial cell and two 1 kg lateral cells digitized at 80 samples per second. An alternative single-piece design — a cruciform flexure instrumented with one Wheatstone bridge per arm, forming a unified three-axis force/torque sensor — is held as a fallback configuration should the discrete three-cell assembly prove difficult to align. The rig is actuator-agnostic, so the magnetic-versus-servo comparison runs entirely on this one fixture under nominally identical thrust conditions; commanded-versus-measured deflection angle (θ, ϕ) is logged across a series of step and ramp commands under the F15-0 thrust condition to extract bandwidth, slew rate, step-response overshoot, steady-state error, and maximum sustained deflection for each actuator. Because each F15-0 firing provides a 3.45 s control window, the independent step/ramp command set for each actuator system is built up across multiple firings on this stand.

Static thrust and materials (jetvane) stand. A single-axis, load-cell-only stand fitted with a steel blast deflector serves two simultaneous purposes: validating the as-fired thrust curve of every motor class used in the program against its published specification, and

holding candidate 3D-printed material coupons directly in the motor exhaust plume to rank flame resistance and surface erosion in support of the Research Question 2 material-zoning conclusions. The deflector and mounting hardware are sized to the F15-0's 3.45 s burn thermal case.

Motor plan and firing counts. Table 3 summarizes the verified motor specifications and the planned firing allocation across flight, both ground stands, and stand commissioning.

Table 3. Motor plan and verified specifications.

Motor	Total impulse	Avg / peak thrust	Burn time	Role
Estes F15-0	49.6 N·s	14.4 N / 25.3 N	3.45 s	Flight, both ground stands, jetvane erosion screen, TVC-balance actuator characterization
Estes/AeroTech E16-4	—	~ 16 N avg	E-class	Stand commissioning and calibration only

Planned firing counts are 16 Estes F15-0 motors (3 flight, 6 on the TVC balance at three firings per actuator system, 5 on the static/materials stand for curve verification and jetvane erosion ranking, and 2 held in reserve) and a minimum of 4 (target 6) Estes/AeroTech E16-4 motors for stand commissioning, at two to three firings per stand.

Calibration and commissioning sequence. Each stand's load cells are first calibrated independently of any motor firing, using a series of known hanging dead weights spanning the expected force range, to establish a force-to-voltage transfer function for every channel. Each stand is then commissioned with a minimum of two low-cost E16-4 motor firings before any data-collection firing is conducted, to validate the as-built stand's measured thrust curve against the motor's independently published reference curve and confirm that structural compliance in the stand itself is not corrupting the force measurement. Only after a stand passes this commissioning check are F15-0 data-collection firings conducted on it. All raw and reduced ground-test data (thrust curves, vector-deflection logs, and materials-erosion observations) are archived under a dedicated data directory structure separating motor thrust-curve data from TVC vector/control data, in the same repository used for flight data release (§5.6).

5.5 Wind Tunnel (Research Questions 3 and 4)

Ground-based aerodynamic characterization is treated as a primary data source for the program, not a secondary check on flight results, since it is the only test environment in

which angle of attack, freestream velocity, and Reynolds number can each be controlled and varied independently — none of which is possible during a single powered flight. A custom open-return, low-speed wind tunnel is therefore constructed in parallel with the flight vehicle, following the modular configuration described by Hofferth (2025), comprising a bell-mouth inlet, a honeycomb-and-screen flow-conditioning section, a converging contraction section sized per the area-ratio and length criteria of Bell and Mehta (1988), an optically accessible test section, and a diffuser leading to a variable-speed fan.

Fan selection and trade study. Because Research Questions 3 and 4 require accurate aerodynamic force measurement rather than maximum achievable airspeed, fan selection is driven primarily by test-section velocity stability and acoustic/vibration noise floor rather than peak flow rate. Two fan classes are traded: a self-contained electronically-commutated 8-inch axial fan with an integrated ten-speed controller, paired with a diffuser-and-fan upgrade to the baseline Hofferth tunnel geometry, and a compact, high-static-pressure 120 mm centrifugal fan adapted to the tunnel inlet through a custom-printed collar, driven by a separate pulse-width-modulated 12 V supply. The first option is selected as the baseline configuration for force-measurement testing on the basis of its purpose-built diffuser pairing and lower acoustic noise at the tunnel's typical operating point; the second is retained as a compact backup path. The complete bill of materials, flow-conditioning screen specification, and full 3D-printed part manifest for the tunnel are maintained in the project's ground-support-equipment documentation and released through the program's public repository (\$5.6).

Instrumentation and calibration. Prior to aerodynamic data collection, the tunnel is characterized across its full operating range using a traversed Pitot-static probe to map test-section freestream velocity and spatial uniformity (target: under 2% RMS variation in the core flow region), following the design and verification practices of Mehta and Bradshaw (1979) and Pope and Harper (1966). Blockage corrections are applied to all measured drag coefficients using the Maskell (1963) bluff-body correction method, and the fan operating point that produces a Reynolds-number similarity ratio matching predicted flight conditions is identified and used as the calibrated test condition for all fin-article testing.

Fin-article force testing. The fabricated fin article is mounted on a deflection-indexed single-fin test mount — either on a strut/sting base or a sidewall half-span mount — permitting controlled 0.5° angle-of-attack increments without remounting the article between runs. Lift and drag forces are recorded at equal angle-of-attack increments from 0° through stall, with a minimum of three repeated runs per angle to assess measurement repeatability; resulting coefficients are validated against the tabulated NACA reference data of Selig (2003) and Selig, Donovan, and Fraser (1989) prior to use in the stability calculation of §4.2.

Computational cross-validation. In parallel with the physical tunnel campaign, a two-dimensional constant-strength vortex-panel method (following Kuethe and Chow, as implemented for each candidate fin cross-section: NACA 0006, NACA 0012, a symmetric double-wedge, and a flat-plate baseline) generates an independent, purely computational

lift-curve and surface-pressure-distribution prediction for every candidate profile, swept through the same 0.5° deflection increments at both the wind-tunnel Reynolds number ($\approx 2 \times 10^5$) and the predicted flight Reynolds number ($\approx 3.4 \times 10^5$). This panel-method baseline is validated internally against thin-airfoil theory (it recovers a lift-curve slope within approximately 10% of the inviscid 2π -per-radian ideal for the NACA 0012 section) before being used as a pre-test prediction against which the physical tunnel measurements are compared; because the panel method is inviscid, it is expected to reproduce lift coefficient and pressure distribution accurately but to systematically miss viscous drag, separation, and stall onset, so the physical tunnel campaign remains the authoritative source for drag coefficient and stall angle while the panel-method results serve as an independent check on the lift-curve slope and as a pre-test prediction for instrumentation sizing. All simulated aerodynamic, trajectory, and control-loop datasets generated during the program — the panel-method polars described here; the RK4-plus-Barrowman trajectory and dispersion simulations of §4.2; the gate-based flight-validation suite and Monte Carlo dispersion runs referenced in §5.6 and §8; and the atmospheric-sweep control-loop simulations of §5.3 — are version-controlled alongside the physical test data in the program's public repository, with each simulation script paired against the dataset it produced, so that every plotted or tabulated simulated result in this paper and its supporting materials can be regenerated and independently checked against the as-built hardware.

5.6 Flight Test Plan, Logging, and Data Sharing (Research Question 5)

Each flight is conducted in compliance with the National Association of Rocketry (2023) safety code and motor classification standards, under the supervision of a NAR-certified or similarly qualified range safety officer. Multiple flights are conducted with the control law configured under each candidate gain set identified in §5.3, and post-flight telemetry recovered from the onboard microSD log is analyzed for peak pitch deviation during the powered phase, gimbal-angle utilization, and qualitative settling/overshoot behavior, to determine which gain configuration produces the most accurate and best-damped attitude tracking. All onboard sensor data (full-rate inertial, barometric, and control-loop telemetry) is logged at the full control-loop sample rate; where the bench/range Wi-Fi telemetry link is in range, a parallel live feed is monitored for real-time anomaly detection, though the onboard log remains the data of record. Upon conclusion of the program, all CAD files, firmware source code, simulation scripts, and the complete flight and ground-test datasets are released through a version-controlled, publicly accessible Git repository, and the principal reduced results from each research question are additionally presented in summary form within this paper.

6. Recovery System

The vehicle's passive fin stability is retained through the coast phase to apogee (predicted apogee ≈ 432 ft at $t \approx 7.0$ s), enabling a conventional dual-event recovery sequence on a RRC3+ flight computer: a primary apogee-detect deployment on the drogue/main channel, with a redundant auxiliary-channel timer-backup deployment armed to fire shortly after

the predicted apogee time should primary detection fail. Both initiators are seated in a single reusable E-Match Mate canister sharing one black-powder charge well. Shock-cord and parachute sizing (1/8" tubular Kevlar cord, 18" ripstop nylon canopy) are verified against the worst-case deployment loading scenario — an off-nominal timer-backup firing while the vehicle retains several meters per second of vertical velocity — yielding a structural safety factor exceeding 800× on the recovery harness and a predicted terminal descent rate near 6 m/s under nominal apogee deployment.

7. Safety and Regulatory Compliance

All flights use a single Estes F15-0 motor (49.6 N·s total impulse, 60 g propellant, F-class), and the fully loaded liftoff mass of 708 g is well under the FAA's Class 1 (model rocket) threshold of 1,500 g loaded weight per motor, requiring no Federal Aviation Administration airworthiness waiver and no NAR/Tripoli high-power certification. Range procedures include remote ignition, a minimum 3 m personnel standoff from both ground-test stands during firing, a fail-safe neutral-gimbal default state on any control-system fault, an independently armed and powered recovery system isolated from the flight-computer power bus, and a remove-before-flight pyrotechnic inhibit on all ejection-charge circuits.

8. Expected Outcomes

The program is expected to produce a quantitative, ground-validated comparison of magnetic-solenoid and servo thrust-vector-control actuators — including bandwidth, slew rate, overshoot, and steady-state error for each — directly informing actuator selection for future closed-loop rocketry programs without requiring a dedicated in-flight A/B comparison. A validated thermally-zoned additive-manufacturing material allocation is expected to demonstrate a measurable dry-mass reduction (approximately 100–150 g, or roughly 15–20% of dry mass) relative to a uniform heat-rated-material baseline, with no corresponding loss of structural margin, offering a transferable design pattern for hobbyist and academic rocketry programs using FDM fabrication. Wind-tunnel-derived aerodynamic coefficients for the flight fin geometry, cross-validated against flight telemetry, will be released as an open dataset characterizing the program's custom low-speed tunnel as a viable, low-cost ground-test instrument for the broader rocketry and STEM-education community. Finally, flight-validated comparison of control-loop gain configurations on a consolidated dual-core single-controller architecture is expected to clarify the practical control-authority and determinism benefits of separating real-time control execution from logging and telemetry I/O on a single low-cost microcontroller, relative to either a single-threaded controller or a distributed multi-board avionics stack. All resulting design files, firmware, simulation code, and flight datasets will be released publicly through the National Association of Rocketry in support of the program's open-source objectives.

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